SYDE 671 · INTRODUCTION TO COMPUTER VISION · ASSIGNMENT 3

[Auto]Stitching Photo Mosaics

Assignment Report · Parts A and B

Homographies, inverse warping, rectification, blended mosaics, and automatic feature matching with RANSAC

Abstract—Part A recovers homographies by least squares from hand-chosen correspondences (`computeH`, a $2n \times 8$ system with $h_{33} = 1$), and warps images with two inverse-warping routines written from scratch: nearest neighbour and bilinear. Both are used to rectify a tower facade and two banners; on a synthetic test with a known answer, bilinear reaches 49.5 dB against 32.8 dB for nearest neighbour, at about 2.3 times the run time. Three mosaics are built from three image sets (a three-photograph clifftop beach set, a river pair and a sea pair), each registered to a reference view and blended with a distance-transform feather and a two-band (low/high frequency) blend. Part B automates the correspondences: Harris corners (28,883 in the reference image) are thinned to 1,500 by adaptive non-maximal suppression, described by blurred, bias/gain-normalised 8×8 patches sampled from a 40×40 window, matched with Lowe's ratio test (0.7), and filtered by a from-scratch 4-point RANSAC (342 and 45 inliers for the two image pairs). Automatic mosaics of all three sets are shown beside the manual ones. The two agree closely; for the river and sea pairs the manual points were found without reference to the automatic result (centre disagreement under 1 px), but for the beach set they were partly screened against it, so that agreement is not an independent test.

Index Terms—homography, direct linear transform, inverse warping, bilinear interpolation, image rectification, feathering, Harris corners, ANMS, MOPS descriptors, RANSAC, image mosaicing.

Scope. The assignment asks for three mosaics in Part A and at least three automatic mosaics in Part B. A course clarification relaxed this to one mosaic (at least two images) for Part A and one (at least three images) for Part B. I did all three anyway, on three image sets: *beach* (IMG_4826, 4827, 4828; three images), *river* (IMG_3361, 3364) and *sea* (IMG_4806, 4807). Each set gets a manual mosaic in Part A and an automatic mosaic in Part B, shown side by side. The beach set is the main example and carries the detailed discussion. All code is written with NumPy and SciPy; none of the banned functions (`cv2.findHomography`, `cv2.warpPerspective`, `cv2.getPerspectiveTransform`, `skimage.transform.warp`, `ProjectiveTransform`) is used. The Harris detector is the supplied `harris.py`.

PART A: IMAGE WARPING AND MOSAICING

A.1 SHOOT THE PICTURES

The mosaic set is three handheld phone photographs taken from a cliff top over a beach: IMG_4826, IMG_4827 and IMG_4828, each downsized to 1200×900. I turned on the spot between shots, so the views are related by a rotation of the camera and a projective transform. Adjacent photographs overlap by roughly half of their width. IMG_4828, the middle view, is the reference frame, which keeps the distortion of the two outer images as small as possible.



Fig. A.1a. The three source photographs. Left to right: IMG_4826, IMG_4827, IMG_4828. The shoreline, beach, and cliff run through all three.

Second and third image sets

A second set is a pair of photographs of a river bank with autumn trees, IMG_3361 and IMG_3364, taken the same way (standing still, turning the camera to the right between shots). The overlap here is large, about 80% of the width, which is more than the 40–70% recommended, so this pair adds little panorama width. I use it to show that the relation is projective. The automatic pipeline of Part B finds 697 matches, and a single homography fits 671 of them (96%) with a mean error of 1.1 px. The inliers are spread over the trees, the water and the stones in the foreground (x from 191 to 1175 px, y from 140 to 807 px). Near and far content therefore move by the same homography, which is what a camera rotating about its centre produces. The fitted homography has a scale of about 1.1–1.2 and a small tilt. The one near object, a dark leaf at the top-left edge of IMG_3361, is the only visible parallax. The same check on the third set, the sea pair IMG_4806 and IMG_4807, gives 701 matches and 645 inliers (92%) with a mean error of 1.3 px, but its inliers lie only in the lower band of the image (the upper half is sea and sky), so a homography fitted to it is less constrained at the top. Both pairs are blended into mosaics in A.4 and B.4.



Fig. A.1b. Second image set: IMG_3361 (left) and IMG_3364 (right). Sixty of the 671 matches that fit one homography are drawn in green; they run in parallel across the whole scene, from the tree tops to the pebbles in the foreground.

The rectification experiments in A.3 use two further photographs: a tower facade (IMG_5185) and a photograph containing two banners (IMG_1456).

> **Caveat on the fixed centre of projection.** The camera was handheld, so it did not rotate exactly about its optical centre. Distant content (sea, beach, far cliff) obeys a single homography well, but the shrubs in the foreground are close to the camera and show parallax. A homography cannot model that, and it appears as slight stretching in the mosaics.

A.2 RECOVER HOMOGRAPHIES

A homography maps a point (x, y) in image 1 to (u, v) in image 2 as $p' \sim Hp$, with H a 3×3 matrix whose lower-right entry is fixed at 1. Multiplying out the projective division gives two linear equations in the eight unknowns $h = (a, b, c, d, e, f, g, h)^T$ for every correspondence:

$$\begin{bmatrix} x & y & 1 & 0 & 0 & 0 & -ux & -uy \\ 0 & 0 & 0 & x & y & 1 & -vx & -vy \end{bmatrix} h = \begin{bmatrix} u \\ v \end{bmatrix}.$$

Stacking n correspondences gives a $2n \times 8$ system $Ah = b$. With $n = 4$ it can be solved exactly, but it is unstable under noise, so I use more points and solve in the least-squares sense with `numpy.linalg.lstsq`.

stitch.py

```
def computeH(im1_pts, im2_pts):
    """Least-squares homography with h33 = 1. Needs n >= 4 points."""
    p1 = np.asarray(im1_pts, float); p2 = np.asarray(im2_pts, float)
    n = len(p1)
    A = np.zeros((2 * n, 8)); b = np.zeros(2 * n)
    x, y = p1[:, 0], p1[:, 1]; u, v = p2[:, 0], p2[:, 1]
    A[0::2, 0:3] = np.c_[x, y, np.ones(n)]
    A[0::2, 6] = -u * x;  A[0::2, 7] = -u * y
    A[1::2, 3:6] = np.c_[x, y, np.ones(n)]
    A[1::2, 6] = -v * x;  A[1::2, 7] = -v * y
    b[0::2] = u; b[1::2] = v
    h, *_ = np.linalg.lstsq(A, b, rcond=None)
    return np.append(h, 1.0).reshape(3, 3)
```

How the correspondences were obtained

I did not click the points with a mouse. They were found in three steps, and the third step matters for how the results should be read.

- **Estimate by eye.** I read distinctive static landmarks (shore rocks, two people on the beach, a beach rock, the cliff groove, a hill clearing) off gridded, zoomed views of both images. Accuracy was about 5–20 px.
- **Refine by correlation.** For each landmark, a 45×45 grey patch around the image-1 point was matched by normalised cross-correlation (`skimage.feature.match_template`) inside a ±18 px window around the eyeballed position in image 2. The best match replaced the estimate. The points were then checked visually by me.
- **Sanity check against the automatic pipeline.** Points that disagreed with the Part B homography were dropped (one hill-clearing match, 7 px off, and one low-correlation rock) and one pairing was corrected. For the six extra 4826 points added to fix conditioning (below), the search window in IMG_4828 was centred on the automatic prediction.

> **Consequence.** The final points agree with the automatic method partly because they were screened against it. The manual-versus-automatic comparison in B.4 is therefore not an independent test, and it is weakest for the 4826 link.

Results

TABLE A.1. CORRESPONDENCES AND RECOVERED HOMOGRAPHIES (IMAGES TO IMG_4828)

Pair	***n***	**System size**	**Residual mean / max**	**Recovered *H***
IMG_4827 → IMG_4828	8	16×8	1.35 / 2.60 px	[[0.4799, 0.0235, 460.84], [−0.2132, 0.8436, 78.04], [−0.0004, 0.0000, 1]]
IMG_4826 → IMG_4828	13	26×8	1.40 / 3.13 px	[[1.718, −0.0151, −786.31], [0.1084, 1.3408, 52.03], [0.0007, −0.0002, 1]]
IMG_3361 → IMG_3364 (river)	10	20×8	0.98 / 2.08 px	[[1.1891, 0.0105, −207.25], [0.0536, 1.0918, −12.87], [0.0002, 0.0000, 1]]
IMG_4806 → IMG_4807 (sea)	9	18×8	1.31 / 4.12 px	[[1.1239, −0.0369, −127.17], [0.0540, 1.0315, −1.43], [0.0001, −0.0001, 1]]

The residual is the distance between Hp_1 and the clicked p_2. The full point tables, the complete $2n \times 8$ matrix A, the vector b, h and H for all four pairs are in `outputs/A2_p4827_4828.txt`, `A2_p4826_4828.txt`, `A2_p3361_3364.txt` and `A2_p4806_4807.txt`. As an example, the first two rows of the 4827 system (shore rock at $(64, 403)$ mapped to $(513, 414)$) are

$$\begin{bmatrix} 64 & 403 & 1 & 0 & 0 & 0 & -32832 & -206739 \\ 0 & 0 & 0 & 64 & 403 & 1 & -26496 & -166842 \end{bmatrix} h = \begin{bmatrix} 513 \\ 414 \end{bmatrix}.$$



Fig. A.2a. The eight correspondences between IMG_4827 (left) and IMG_4828 (right), numbered. Points 2–4 are the two people and a rock on the beach; 6–8 are on the cliff.



Fig. A.2b. The thirteen correspondences between IMG_4826 (left) and IMG_4828 (right).



Fig. A.2c. The ten correspondences between IMG_3361 (left) and IMG_3364 (right) for the river set: tree-top and bush landmarks, the shore notch, stones in the river, and stones on the near bank.



Fig. A.2d. The nine correspondences between IMG_4806 (left) and IMG_4807 (right) for the sea set: two shore rocks, three bush and flower-bush landmarks, and three points on clouds.

Correspondences for the river and sea sets

For these two pairs I changed the procedure so that the result could be compared independently with Part B. I chose landmarks by eye in the first image (14 for river, 13 for sea). Each was then located in the second image by normalised cross-correlation of a 51×51 grey patch against the *whole* second image at scales from 0.90 to 1.40 (the second photograph is slightly zoomed; the scaling uses my own bilinear warp), so no estimate of the answer was needed and the automatic homography played no part. I then looked at the candidates on the images and kept those that clearly sat on the same feature (10 for river, 9 for sea; script `estimate_points.py`). Rejected candidates were sky-only patches, which match almost anywhere (correlation 0.98 with an equally good runner-up), patches with correlation below 0.6, and ambiguous rock and reef patches that landed in the sky. Two cloud points in the sea set were accepted on visual agreement, because correlation along a smooth cloud streak has a high runner-up on the streak itself. Clouds can drift between shots, so these are the weakest points.

A conditioning problem and its fix

My first 4826 → 4828 fit used seven points and had a residual of about 2 px, which looks fine. But all seven lay in one patch of roughly 190×260 px, and the fitted homography sent the image corners thousands of pixels away from where the automatic homography (45 inliers) sends them. A small residual only says that H is consistent at the points; it says nothing about extrapolation. Adding six points across the overlap (the hill-clearing edges, a boulder pair, a boulder, a dark rock, a tidal pool) fixed this: the corners now land 3–51 px from the automatic result, while the residual stays at 1.4 px.

The 4827 homography was well conditioned from the start. Compared with the automatic one, the two agree to 0.7 px at the image centre and to 5 px at the left corners, but they differ by 47–57 px at the far right corners, which are extrapolated beyond the overlap.

A.3 WARP THE IMAGES

Both warping functions use inverse warping: for each pixel of the output canvas, the inverse homography gives a source position, and the output takes its value from there. This leaves no holes. The output canvas is found by sending the four image corners through H and taking the integer bounding box. The convention is that $(0, 0)$ is the centre of the first pixel, so $(0.5, 0)$ lies halfway between the first and second pixels.

stitch.py

```
def warpImageNearestNeighbor(im, H, bbox=None):
    h, w = im.shape[:2]
    bbox = bbox or warp_bbox(im.shape, H)
    sx, sy = _inverse_coords(im.shape, H, bbox)
    xi, yi = np.rint(sx).astype(int), np.rint(sy).astype(int)
    valid = (xi >= 0) & (xi < w) & (yi >= 0) & (yi < h)
    out = np.zeros(sx.shape + im.shape[2:])
    out[valid] = im[yi[valid], xi[valid]]
    return out, valid.astype(float), (bbox[0], bbox[1])

def warpImageBilinear(im, H, bbox=None):
    h, w = im.shape[:2]
    bbox = bbox or warp_bbox(im.shape, H)
    sx, sy = _inverse_coords(im.shape, H, bbox)
    valid = (sx >= 0) & (sx <= w - 1) & (sy >= 0) & (sy <= h - 1)
    x0 = np.clip(np.floor(sx).astype(int), 0, w - 2)
    y0 = np.clip(np.floor(sy).astype(int), 0, h - 2)
    dx = np.clip(sx - x0, 0, 1)[..., None]; dy = np.clip(sy - y0, 0, 1)[..., None]
    out = ((1 - dx) * (1 - dy) * im[y0, x0] + dx * (1 - dy) * im[y0, x0 + 1]
           + (1 - dx) * dy * im[y0 + 1, x0] + dx * dy * im[y0 + 1, x0 + 1])
    out[~valid] = 0
    return out, valid.astype(float), (bbox[0], bbox[1])
```

Each function returns the warped image, an alpha mask (1 where the output pixel came from inside the source image and 0 where it is empty), and the canvas offset. The alpha mask is what the blending in A.4 uses to ignore empty regions. Neither function calls a library interpolator; the four-neighbour weights are written out explicitly. (The only library interpolation in the project is `scipy.ndimage.map_coordinates`, used when sampling descriptors in B.2, which is not part of the warping.)

Rectification

To test the code, I rectified three known rectangles: a tower facade, and two banners that both appear in IMG_1456. For each, the four corners of the object in the photograph were mapped to an upright rectangle with `computeH`. The corners were read off zoomed, gridded views of the photographs. The target sizes (300×600 for the tower, 300×700 for each banner) are my assumptions about the proportions of the real objects; I did not measure them, so the rectified shapes are plausible but not metrically exact.

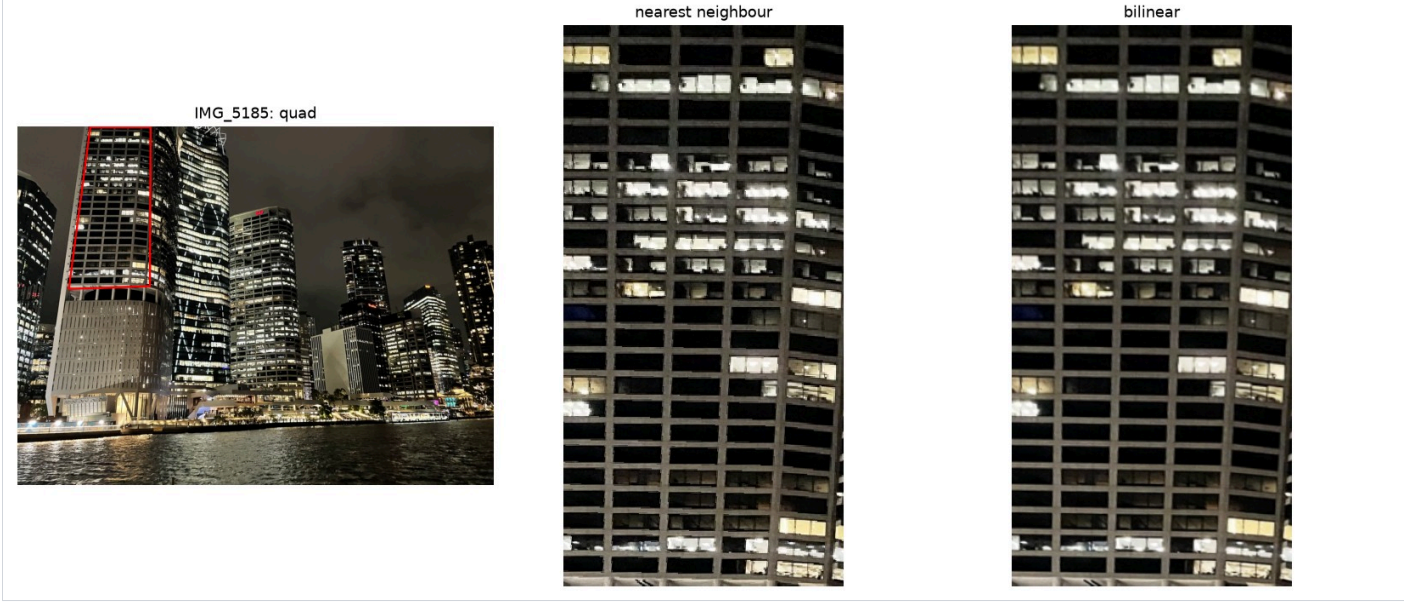


Fig. A.3a. Tower facade (IMG_5185). Left: the source photograph with the chosen quadrilateral. Centre: nearest-neighbour result. Right: bilinear result. The window rows become level. The right edge stays slightly curved because the building itself is curved, which no homography can remove.



Fig. A.3b. First banner (IMG_1456). The converging edges become parallel and the text lines become horizontal.



Fig. A.3c. Second banner (IMG_1456), warped with its own homography from the same photograph.

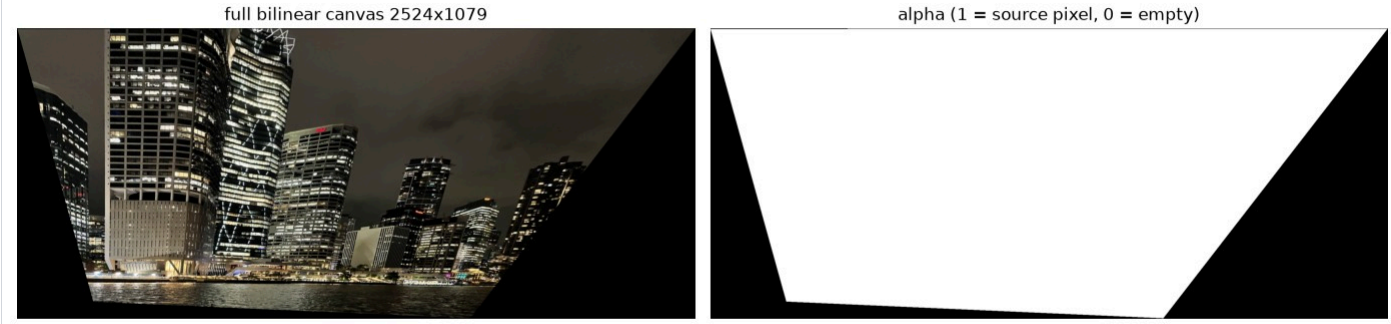


Fig. A.3d. The full warped canvas for the tower (2524×1079, from the four transformed corners) and its alpha mask. Black pixels in the image are empty (alpha 0). For the banners the full canvases are about 2900×2900 and 3500×3000 px because the rectifying homography stretches the image far outside the object, so only the cropped rectangles are shown.

Nearest neighbour versus bilinear

Rendering the output at 4× makes the difference visible. Nearest neighbour copies the single closest pixel, so edges become staircases and thin lines break up. Bilinear mixes the four surrounding pixels, so edges are smooth, at the cost of a small amount of softening.

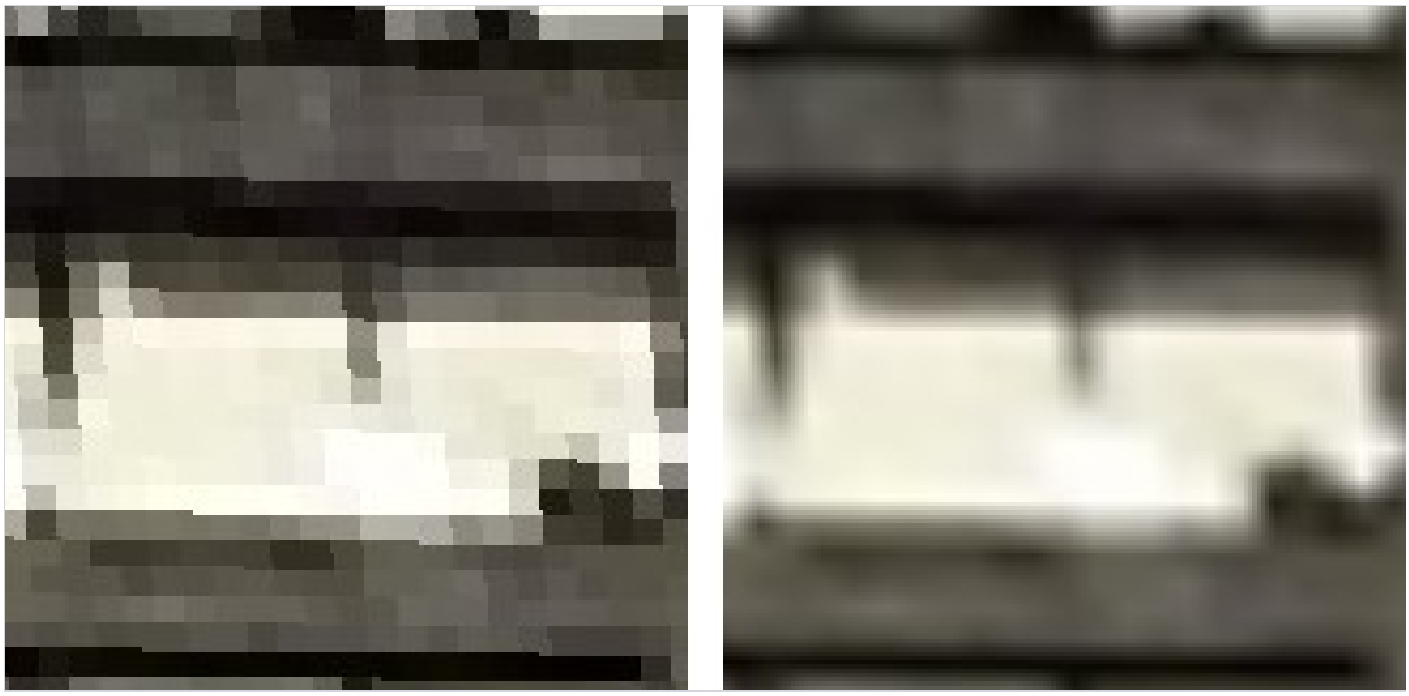


Fig. A.3e. A 160×160 crop of the tower rendered at 4×. Left: nearest neighbour (blocky steps along edges). Right: bilinear (smooth).



Fig. A.3f. The same comparison for the first banner. The letters show the same blocky versus smooth difference.

To measure quality rather than only look at it, I built a test with a known answer. The plane texture is an analytic function $f(u, v)$ (a sum of three sinusoids). The "photograph" is f sampled exactly through the tower homography, with no interpolation anywhere. Rectifying it back with each method and comparing against f on the 180,000 valid pixels gives:

TABLE A.2. ACCURACY AND SPEED OF THE TWO INTERPOLATION METHODS

Method	RMSE	PSNR	300×700 output	1200×2800 output	Typical slow-down
Nearest neighbour	0.0228	32.8 dB	16–20 ms	355 ms	1×
Bilinear	0.0034	49.5 dB	31–48 ms	690–750 ms	about 2.3× (range 1.6–3.6×)

Timings are the best of five runs on the banner homographies at three output scales; the full list is in `outputs/A3_report.txt`.

Trade-off. Bilinear is roughly twice as slow as nearest neighbour but about 16 dB better in PSNR. Nearest neighbour does one rounded lookup per pixel and one array read; bilinear does four reads and a weighted sum of them. Nearest neighbour is acceptable for quick previews or when the output is a downsample, and bilinear is the better choice for a final image. In a mosaic the extra cost is small next to the rest of the pipeline, so I use bilinear throughout.

A metric I rejected. My first accuracy test warped an image forward and back and compared with the original. Nearest neighbour scored "infinite" PSNR, because the round trip is an upsample followed by a downsample that returns the very same pixels. That number said nothing about interpolation quality, so I replaced it with the analytic test above.

A.4 BLEND THE IMAGES INTO A MOSAIC

The mosaic uses all three photographs, with IMG_4828 as the reference. I use the three images because they give a wider panorama than two, and the same set is required for Part B. The procedure is:

- **Homographies.** H_{27} (4827 → 4828) and H_{26} (4826 → 4828) come from the A.2 correspondences; 4828 maps by the identity.
- **Canvas.** The four corners of each image are sent through its homography; the canvas is the union of the three bounding boxes (3243×1987 px).
- **Warp.** Each image is inverse-warped into the shared canvas with `warpImageBilinear`, giving one layer and one alpha mask per image.
- **Weights.** Each layer's weight is the Euclidean distance transform of its alpha mask, normalised to 1. It is largest deep inside the image and falls linearly to 0 at the image border, which is the "distance transform" suggestion in the assignment. At every canvas pixel the weights of the layers that cover it are normalised to sum to 1.
- **Feathering (1 band).** The output is the weighted average of the layers. This alone removes hard seams, but misaligned content in the overlap is averaged into a blur (ghosting).
- **Two-band blend.** Each layer is split into a low-pass part (Gaussian, $\sigma = 15$) and a high-pass remainder. The low-pass parts are combined with the feather weights, which gives a smooth exposure transition. The high-pass parts, which carry the edges and texture, are taken from the image with the largest weight at each pixel, so they are not averaged. A hard choice produced thin lines where the winner changes, so the hand-over is softened by blurring the winner masks ($\sigma = 2.5$ px).



Fig. A.4a. The manual mosaic of IMG_4826, IMG_4827 and IMG_4828 (two-band blend). The black areas are canvas that no photograph covers; they come from the camera tilting between shots.



Fig. A.4b. A crop of the 4827/4828 overlap in the foreground vegetation. Left: feathering only. Right: the two-band blend. The shrubs are visibly blurrier on the left, because the foreground has parallax and the averaged images do not line up. The two-band version keeps their detail.

On the sky and the far shoreline the two methods look the same: the exposures of the three photographs already match, and distant content is well aligned, so a wide feather is enough there. The two-band blend matters in the near-field texture, where it avoids averaging misaligned pixels. The mean absolute difference between the two mosaics is only about 1.3 grey levels out of 255, so the benefit is local rather than global.

Two more mosaics: river and sea

The same procedure was applied to the two pairs, with the second image of each pair as the reference (so only the first image is warped) and the manual homographies of Table A.1. Because the overlap is about 80%, the mosaics are only slightly larger than one photograph; the point here is the registration and the blend, not panorama width.



Fig. A.4c. Manual mosaic of IMG_3361 and IMG_3364 (river), two-band blend. The trees, the shoreline and the stones line up across the join; the near-camera leaf at the top left is the only content that is not far from the camera.



Fig. A.4d. Manual mosaic of IMG_4806 and IMG_4807 (sea), two-band blend. The shoreline, the reef and the horizon are continuous.



Fig. A.4e. Feather only (left) against the two-band blend (right) for the river set (top, stones on the near bank) and the sea set (bottom, bushes). The two-band result is visibly sharper in both.

Here the two blends differ clearly, more than for the beach set. The mean absolute difference between the feather-only and two-band mosaics is 4.4 grey levels for river and 3.1 for sea (against 1.3 for beach), and the feather-only versions are softer: stones, pebbles and leaves lose their edges. A likely reason is that with an 80% overlap most of the canvas is an average of two images, and the warped first image is magnified by about 1.1–1.2 and so is softer than the reference, while any small misregistration also blurs an average. The two-band blend takes the fine detail from one image per pixel and averages only the low frequencies, so it keeps the sharpness. A very faint seam can still be found in the smooth sky of both mosaics if one looks for it (in the river mosaic near the top right, in the sea mosaic near the top).

Remaining artifacts

- A faint diagonal line is visible in the sky at the left boundary of IMG_4826. It appears in both the feather-only and the two-band result, so it is not caused by the high-frequency selection. I suspect a slight exposure or vignetting difference at that image's edge, which a per-image gain correction would remove. I did not add one.
- The foreground shrubs are slightly stretched, and a few edges in them do not line up, because of parallax.

PART B: FEATURE MATCHING FOR AUTOSTITCHING

B.1 HARRIS CORNERS AND ANMS

I used the supplied Harris detector (single scale, 20 px border discarded). On IMG_4828 it returns 28,883 local maxima. Most of them are weak maxima in textureless sky, which are useless for matching and would swamp the later stages.

Adaptive non-maximal suppression (Brown et al., Section 3) keeps points that are both strong and well spread out. For each corner i with strength f_i, its suppression radius is the distance to the nearest corner that is clearly stronger:

$$r_i = \min_j \|x_i - x_j\| \quad \text{subject to} \quad f_i < c_{\text{robust}}\, f_j, \qquad c_{\text{robust}} = 0.9,$$

and the 1,500 points with the largest radii are kept. Because this needs a distance matrix over all candidate pairs, I only consider the 4,000 strongest corners; with all 28,883 the matrix does not fit in practical memory and the first version hung.

stitch.py

```
def anms(pts, strengths, n_keep=500, c_robust=0.9, max_candidates=4000):
    top = np.argsort(-strengths)[:max_candidates]
    pts, strengths = pts[top], strengths[top]
    d2 = dist2(pts.astype(float), pts.astype(float))
    stronger = strengths[:, None] < c_robust * strengths[None, :]
    d2 = np.where(stronger, d2, np.inf)
    radii = d2.min(axis=1)
    keep = np.argsort(-radii)[:n_keep]
    return pts[keep], radii[keep]
```

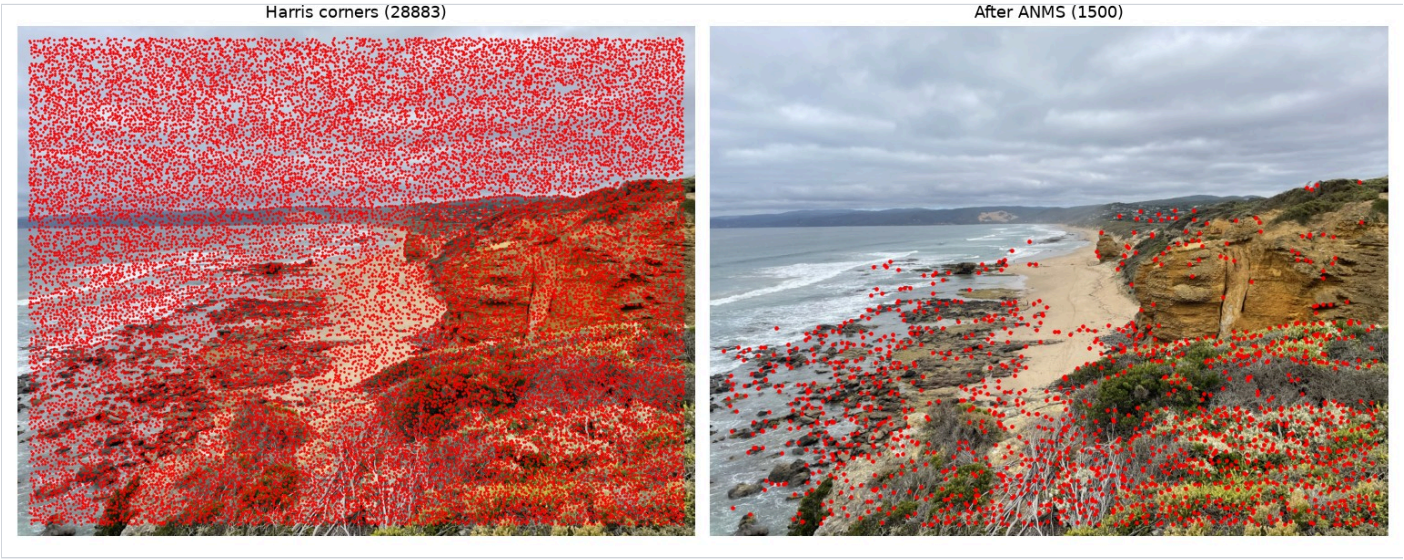


Fig. B.1. IMG_4828. Left: all 28,883 Harris corners; the sky is covered in weak maxima. Right: the 1,500 points kept by ANMS, which sit on real texture (cliff, rocks, shrubs) and are spread out instead of clumped.

ANMS leaves the sky and open sea nearly empty, because their corner response is weak, and it places many points on the foreground shrubs. That is a poor place for matching, since the shrubs are the content with the most parallax (see B.3).

B.2 FEATURE DESCRIPTORS

For each kept corner I take a 40×40 window from the grey image, blurred with a Gaussian ($\sigma = 2$), and sample an 8×8 grid from it with a spacing of 5 px (bilinear sampling, axis-aligned, no rotation). Sampling from the blurred larger window makes the descriptor insensitive to small localisation errors. Each descriptor is then bias/gain normalised, $\hat{d} = (d - \mu)/\sigma$, so that it is unchanged by a uniform brightness offset and contrast scale. This gives a 64-vector per point.

stitch.py

```
def extract_descriptors(im, pts, window=40, size=8):
    g = gaussian_filter(to_gray(im), 2)
    step = window // size
    offs = (np.arange(size) - size / 2 + 0.5) * step
    oy, ox = np.meshgrid(offs, offs, indexing='ij')
    desc = []
    for x, y in pts:
        p = map_coordinates(g, [oy + y, ox + x], order=1, mode='nearest')
        desc.append(((p - p.mean()) / (p.std() + 1e-8)).ravel())
    return np.array(desc)
```

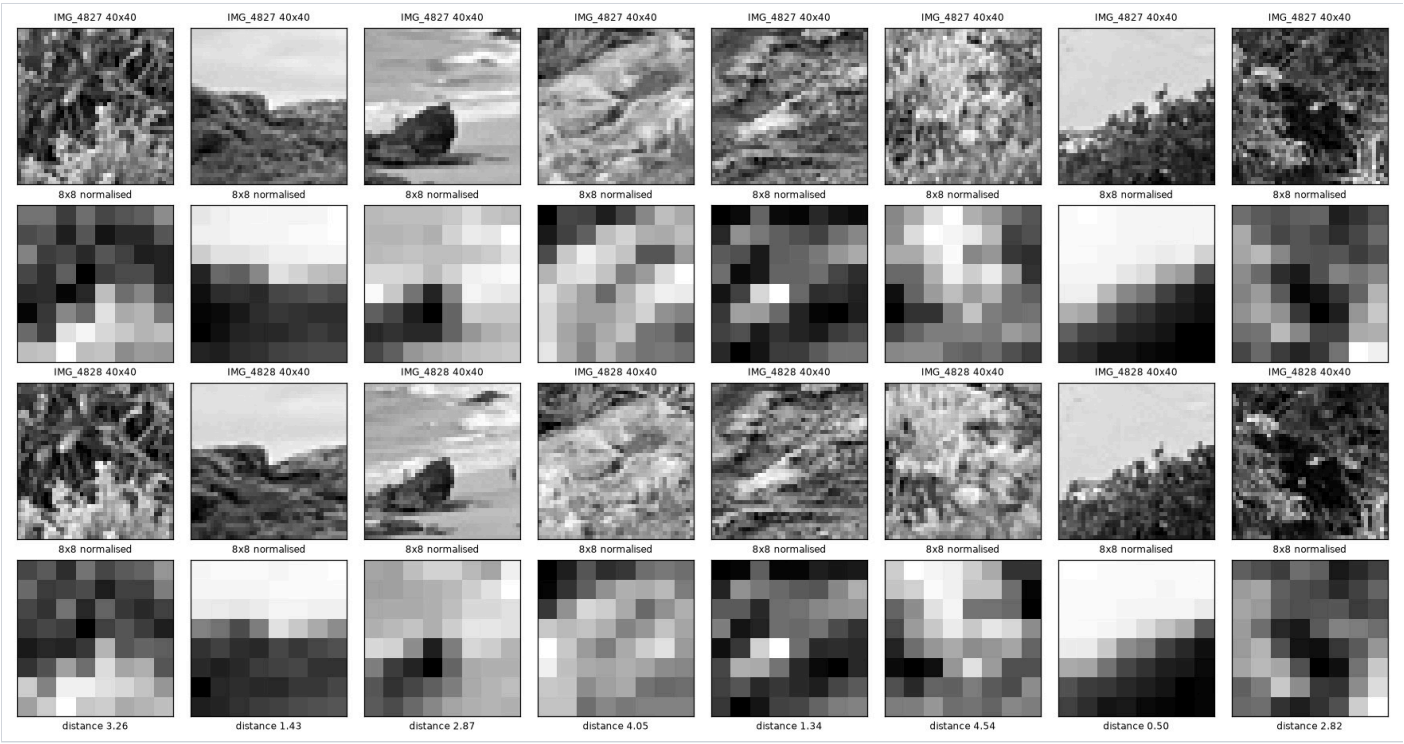


Fig. B.2. Eight spread-out landmarks that RANSAC accepted as correct matches between IMG_4827 and IMG_4828 (chosen by farthest-point sampling). Row 1: the 40×40 grey window in IMG_4827. Row 2: its normalised 8×8 descriptor, displayed with its own contrast. Rows 3 and 4: the same for the matching point in IMG_4828, with the Euclidean distance between the two descriptors under it. Each pair looks alike despite the change of viewpoint.

Two checks support the descriptors. First, bias/gain normalisation works as claimed: applying $I \to 0.6\,I + 0.15$ to the whole image changes the 200 test descriptors by at most 5×10^{-7} (the small residual comes from the 10^{-8} added to the standard deviation), and every descriptor has mean 0 and standard deviation 1. Second, the descriptors discriminate: the eight matched pairs in Fig. B.2 are on average 2.6 apart, against 11.3 for randomly paired descriptors. The matched pairs were selected because they passed the ratio test and RANSAC, so the first number is optimistic; it shows that correct matches are close, not how often a close pair is correct. The check is in `outputs/B2_report.txt`.

B.3 FEATURE MATCHING

Each descriptor in one image is compared with all descriptors in the other by Euclidean distance. Following Lowe, a match is accepted only if the distance to the best neighbour is below a fixed fraction of the distance to the second-best neighbour. This rejects ambiguous features (repeated textures such as waves and foliage), whose two best matches are almost equally good. I tried three thresholds (Table B.0) and use 0.7. At 0.6 the weak 4826 link keeps only 12 matches, too few for a stable fit across the overlap. At 0.8 there are more

inliers in absolute terms, but the share of rejected matches rises (16% against 10% for 4826 and 9% against 5% for 4827), so 0.7 is a compromise between quantity and purity rather than a sharp optimum.

TABLE B.0. EFFECT OF THE RATIO THRESHOLD (MATCHES / RANSAC INLIERS)

Pair	0.6	0.7 (used)	0.8
IMG_4826 → IMG_4828	12 / 12	50 / 45	97 / 81
IMG_4827 → IMG_4828	248 / 241	359 / 342	473 / 430

stitch.py

```
def match_features(d1, d2, ratio=0.6):
    D = dist2(d1, d2)
    order = np.argsort(D, axis=1)
    i = np.arange(len(d1))
    nn1, nn2 = D[i, order[:, 0]], D[i, order[:, 1]]
    ok = np.sqrt(np.maximum(nn1, 0)) < ratio * np.sqrt(np.maximum(nn2, 0))
    return np.c_[i[ok], order[ok, 0]]
```

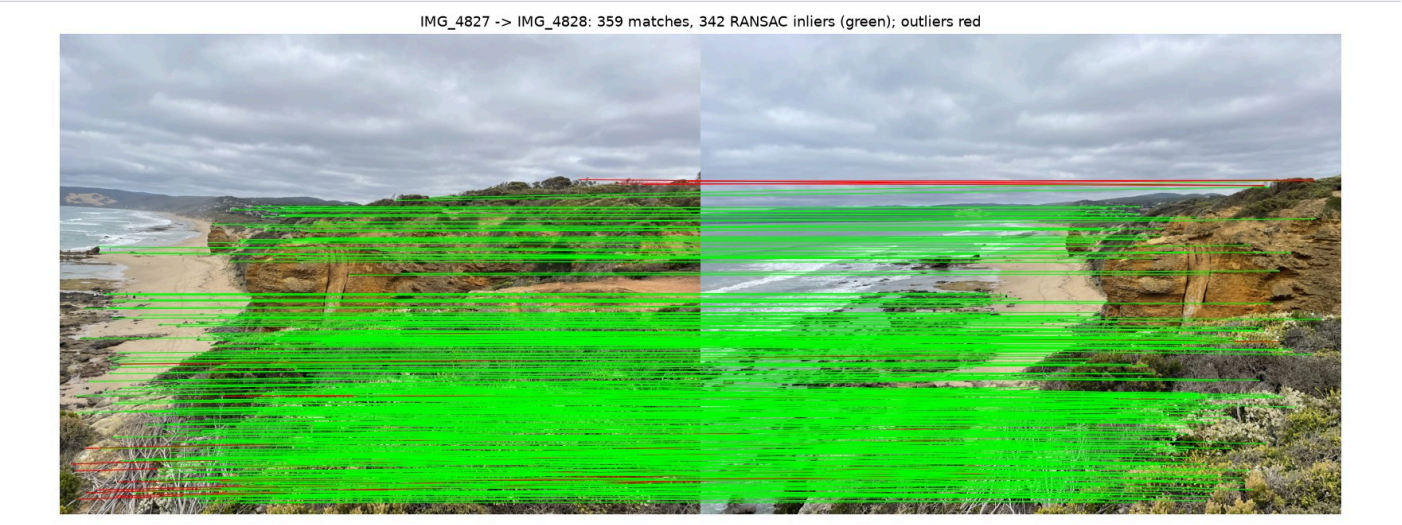


Fig. B.3a. IMG_4827 → IMG_4828: 359 ratio-test matches, of which RANSAC keeps 342 (green) and rejects 17 (red).

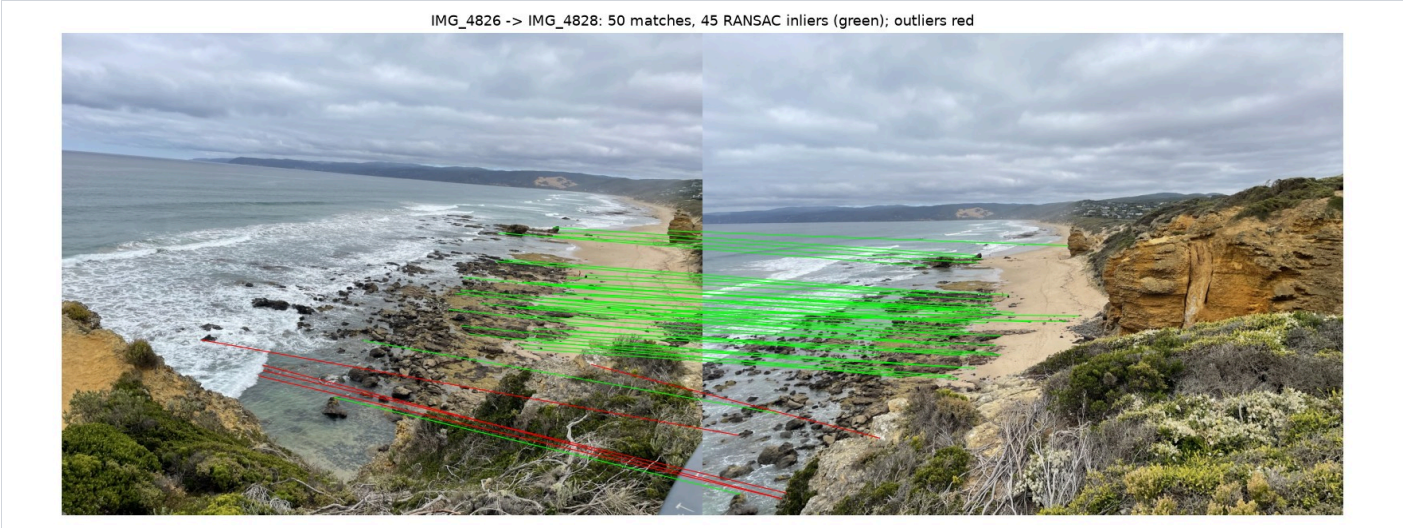


Fig. B.3b. IMG_4826 → IMG_4828: only 50 matches, of which 45 are RANSAC inliers. The matches sit on the beach and rocks in the middle distance. The red outliers are in the foreground shrubs, where the repeated texture and parallax confuse the descriptors.

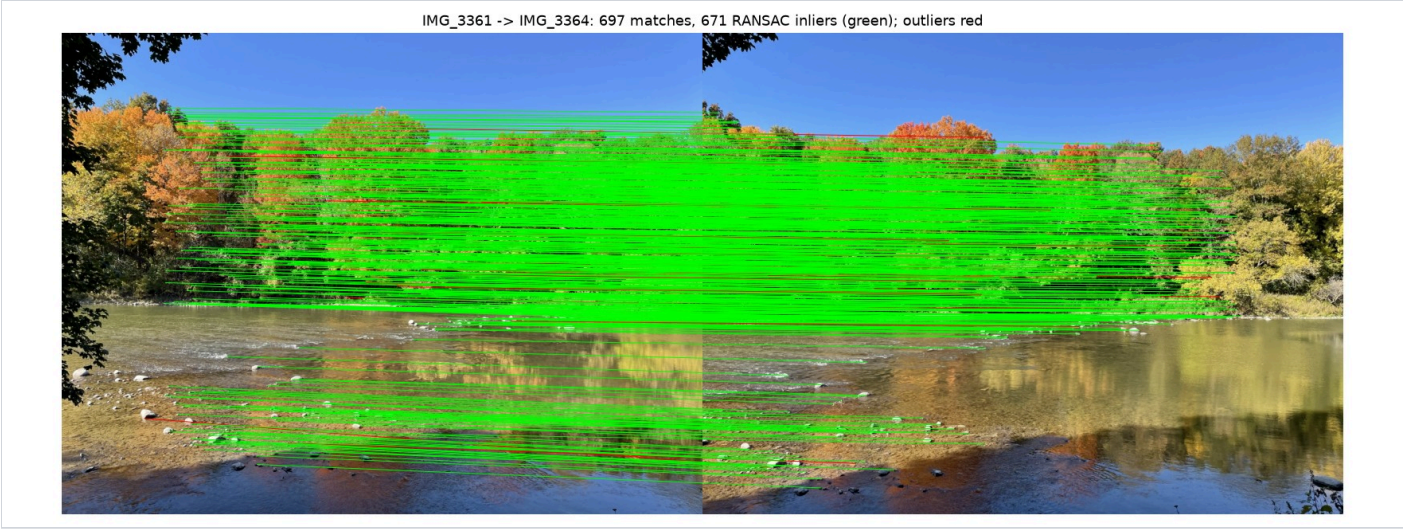


Fig. B.3c. IMG_3361 → IMG_3364 (river): 697 ratio-test matches, 671 RANSAC inliers (green), 26 outliers (red).

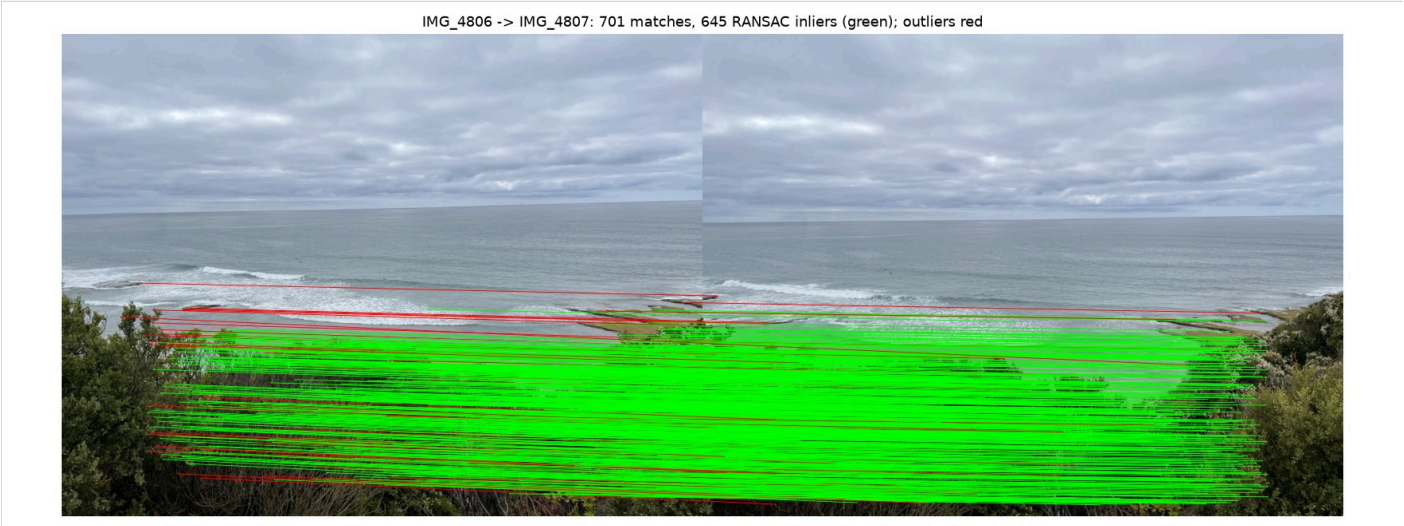


Fig. B.3d. IMG_4806 → IMG_4807 (sea): 701 matches, 645 inliers, 56 outliers. The inliers are in the lower band, on the shoreline and bushes; the sea and sky above contribute almost none.

The descriptors are not scale- or rotation-invariant, so they work only because the three photographs share a focal length and orientation. Pairs with a zoom change would not match.

B.4 RANSAC AND AUTOMATIC MOSAIC

Some of the ratio-test matches are still wrong, and a least-squares fit is pulled far off by even a few of them. 4-point RANSAC fits the homography to random minimal samples and keeps the model that most matches agree with: repeat 5,000 times, pick four random matches, compute H with `computeH`, count the matches whose reprojection error is below $\varepsilon = 3$ px, and keep the largest consensus set. The final homography is refit by least squares on all inliers of the best model.

stitch.py

```
def ransac_homography(p1, p2, n_iter=5000, eps=3.0, seed=0):
    rng = np.random.default_rng(seed)
    best = np.zeros(len(p1), bool)
    for _ in range(n_iter):
        idx = rng.choice(len(p1), 4, replace=False)
        H = computeH(p1[idx], p2[idx])
        if not np.all(np.isfinite(H)):
            continue
        err = np.linalg.norm(apply_H(H, p1) - p2, axis=1)
        inl = err < eps
        if inl.sum() > best.sum():
            best = inl
    return computeH(p1[best], p2[best]), best
```

TABLE B.1. AUTOMATIC PIPELINE RESULTS (1,500 ANMS POINTS PER IMAGE, RATIO 0.7, E = 3 PX)

Pair	Ratio-test matches	RANSAC inliers	Automatic H
IMG_4827 → IMG_4828	359	342	[[0.4759, 0.0085, 461.90], [−0.2100, 0.8202, 82.82], [−0.0004, −0.00002, 1]]
IMG_4826 → IMG_4828	50	45	[[1.7756, −0.0134, −812.62], [0.1263, 1.3828, 40.59], [0.0007, −0.0002, 1]]
IMG_3361 → IMG_3364 (river)	697	671	[[1.1871, 0.0072, −206.19], [0.0520, 1.0857, −10.46], [0.0002, 0.0000, 1]]
IMG_4806 → IMG_4807 (sea)	701	645	[[1.1405, −0.0324, −135.49], [0.0601, 1.0491, −9.24], [0.0001, −0.00005, 1]]

Both automatic homographies were then used with the same blending code as in A.4 (three images, 4828 as the reference, two-band blend), so the only difference between the two mosaics is where the homographies come from.

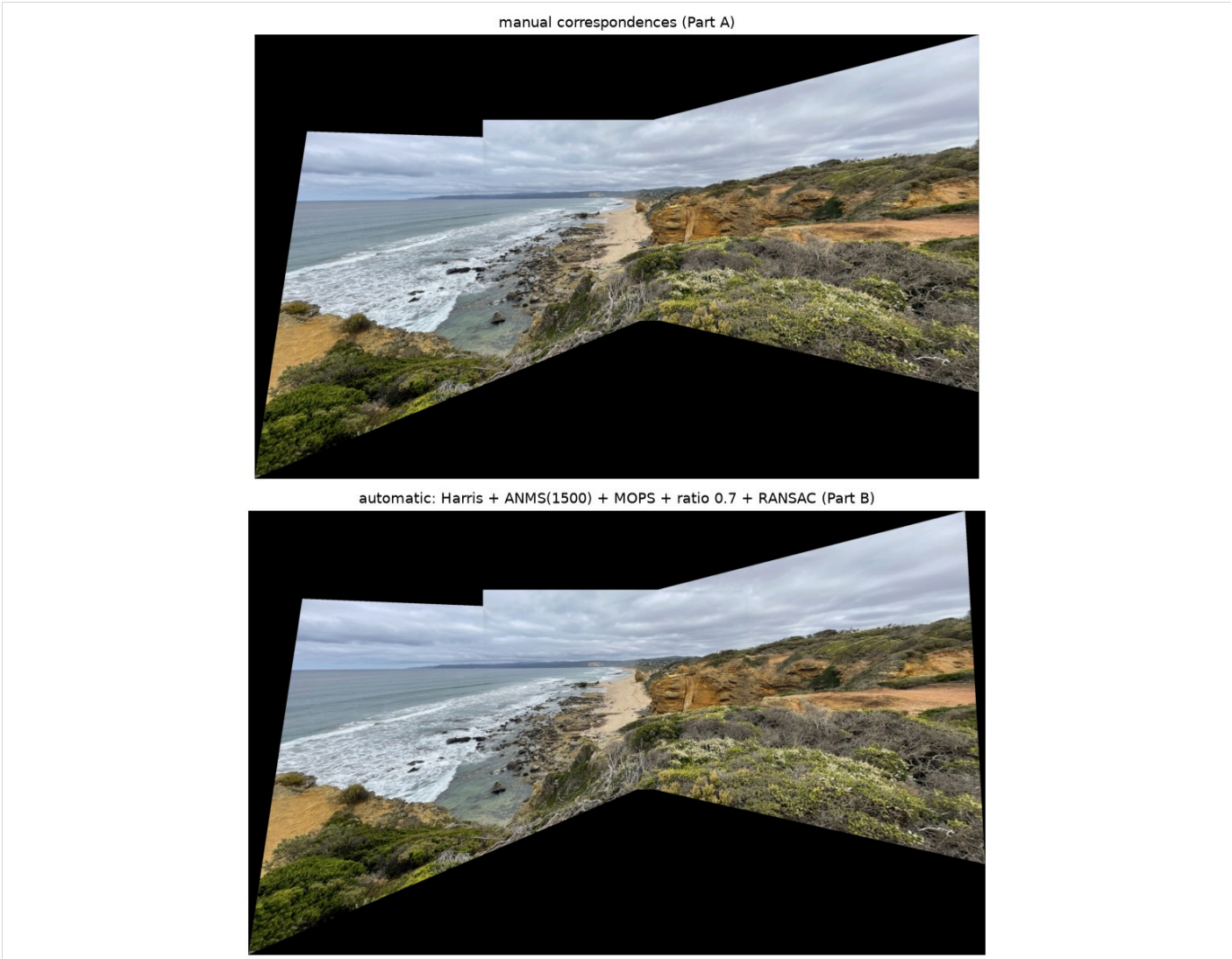


Fig. B.4a. Top: the manual mosaic from Part A (hand-chosen correspondences). Bottom: the automatic mosaic (Harris, ANMS, MOPS descriptors, ratio test, RANSAC). The two are almost identical in layout and are equally seamless; they differ by a few pixels along the outer edges of the canvas, as Table B.2 predicts.

Automatic mosaics: river and sea



Fig. B.4b. River set. Top: manual mosaic (Part A). Bottom: automatic mosaic (Part B). They are visually indistinguishable.



Fig. B.4c. Sea set. Top: manual. Bottom: automatic. The two differ by up to about 11 px at the top-left corner of the canvas (Table B.2).

Manual versus automatic

TABLE B.2. DISAGREEMENT BETWEEN THE MANUAL AND AUTOMATIC HOMOGRAPHIES, IN PIXELS, AFTER MAPPING POINTS INTO IMG_4828

Pair	Image centre	Top left	Top right	Bottom right	Bottom left
IMG_4827 → IMG_4828	0.7	4.9	56.5	46.6	5.3
IMG_4826 → IMG_4828	2.4	28.7	2.6	5.6	51.2
IMG_3361 → IMG_3364 (river)	0.5	2.6	0.8	0.9	3.0
IMG_4806 → IMG_4807 (sea)	0.8	11.4	2.6	3.1	3.1

At the image centres, where the correspondences are, the two methods agree to 1–2 px. The disagreement grows at the corners that lie farthest from the overlap, where each homography is an extrapolation; in the mosaic those are the outer edges, which is why the canvas outline differs slightly. Because the manual points were screened against the automatic result (A.2), this agreement is expected and should not be read as proof that the manual points are independently correct. Where the two do differ, I cannot say from this data which is closer to the truth.

The river and sea rows are the more informative ones. Their manual points were found by correlation over the whole second image and never compared with the automatic homography, so agreement here is genuine evidence that both methods recover the same transform. The two agree to under 1 px at the centre and to within 3 px at the corners for river, and to within 3 px at three sea corners and 11 px at the top left (where only cloud points constrain the manual fit and clouds can drift). With a large overlap and texture across the image these two pairs are easy and well conditioned, which is why the agreement is so close; the beach set, with its smaller overlap and parallax, is the harder test.

The advantage of the automatic pipeline is that it uses hundreds of points spread over the overlap (342 for the 4827 link) with no manual work, and it needs no judgement about which features are reliable. Its weakness is the 4826 link, where only 45 inliers survive and all lie in the middle distance, so the extrapolation to the far corners is the least certain part of that homography.

LIMITATIONS

- **Manual points.** None of the correspondences were mouse-clicked. For the beach set they were estimated by eye, refined by correlation, and screened against the automatic homography (A.2), so that comparison is not independent. For the river and sea sets they were located by whole-image correlation without reference to the automatic result, so those comparisons are independent, but the landmarks and the accept/reject decisions were still mine. `pick_points.py` is included for clicking points by hand.
- **Parallax.** The camera was handheld and the foreground is close, so a single homography cannot align the shrubs. The two-band blend hides most of the resulting ghosting but not the stretching.
- **Rectification sizes are assumed.** The output proportions of the tower and banners (1:2 and 3:7) are plausible guesses, not measurements, and the tower is curved.
- **Descriptors are not scale or rotation invariant**, and the 4826 link has only 45 inliers.
- **Two of the three mosaics are easy.** The river and sea pairs overlap by about 80%, so their mosaics are barely larger than a single photograph and say more about registration than about panoramas. The beach set is the demanding one.
- **Exposure.** A faint seam remains at the left edge of IMG_4826 and, very faintly, in the sky of the river and sea mosaics; no gain compensation is applied.

WHAT I LEARNED

A small fitting residual does not mean a good homography. Seven well-fitting points in one corner of the image gave a homography that was off by thousands of pixels at the image corners, and only spreading the points across the overlap fixed it. Residuals at the data points must be paired with a check where the transform will actually be used.

Evaluating interpolation needs a ground truth. My first metric looked precise and meant nothing, while a synthetic texture with a known answer gave a clear 16 dB gap between the methods. Blending also showed that feathering is enough where images agree, and that the extra frequency band earns its place only where they do not.

Finally, ANMS and the ratio test do different jobs. ANMS decides where to look, and removes the thousands of noise corners in the sky; the ratio test decides which matches to trust, and removes ambiguous texture; RANSAC then removes what is left. The weak 4826 link shows what happens when each of those has little to work with.

FILES, SOURCES, AND CREDITS

Run `python a2.py`, `python a3.py`, `python run_all.py` and `python report_figs.py` in that order from this folder, in an environment with the packages in `requirements.txt`. Figures are written to `outputs/`; the web copies on this page are in `media/`.

- `stitch.py`: `computeH`, `apply_H`, `warpImageNearestNeighbor`, `warpImageBilinear`, `blend`, `harris`, `anms`, `extract_descriptors`, `match_features`, `ransac_homography`.
- `a2.py` (A.2 tables and figures), `a3.py` (rectification and the interpolation study), `run_all.py` (A.4 and Part B), `report_figs.py` (extra report figures), `estimate_points.py` (correlation search for the river and sea correspondences), `pick_points.py` (point-clicking tool).
- `harris.py`: sample Harris code supplied with the assignment, unmodified.
- `points/*.json`: the correspondences; `A2_notes.md`: how they were obtained; `outputs/`: figures and text reports.

1. M. Brown, R. Szeliski, S. Winder, "Multi-Image Matching using Multi-Scale Oriented Patches", CVPR 2005.
2. D. G. Lowe, "Distinctive Image Features from Scale-Invariant Keypoints", IJCV 2004 (ratio test).
3. A. R. Smith, "A Pixel Is Not a Little Square" (pixel-centre convention).

All photographs were taken by me.